

“TOUCHSTAR”: Exploring Generative AI-Powered Interactive Narrative in Cinematic Virtual Reality

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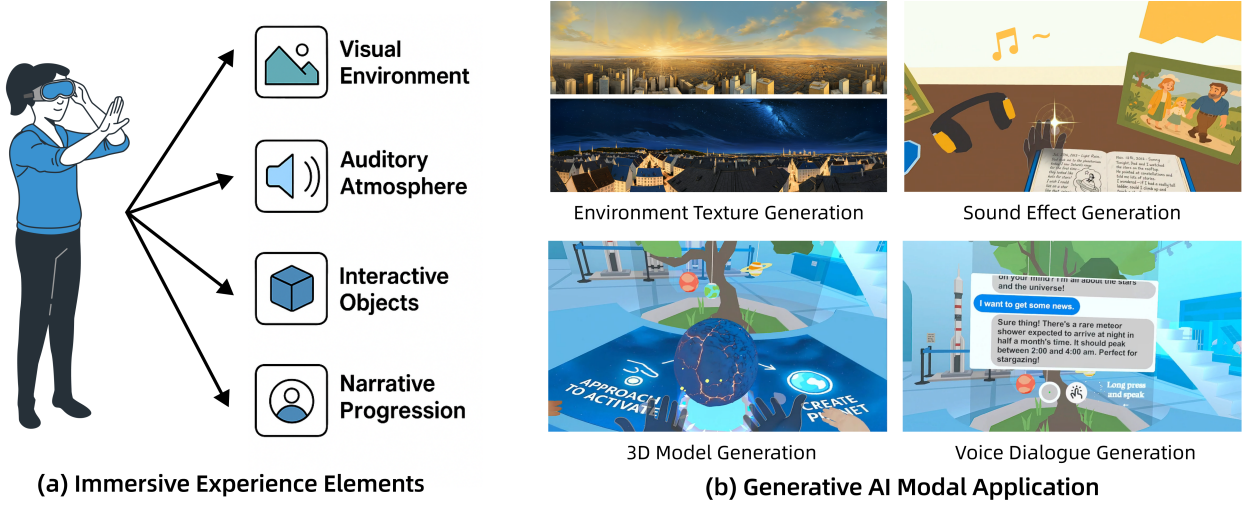


Figure 1: Generative AI-Powered VR Interactive Narrative Design: (a) Key Elements of Immersive Experience in VR. (b) Corresponding Applications of Generative AI Modalities.

ABSTRACT

TouchStar is an immersive VR interactive film integrating generative artificial intelligence technologies. It tells the story of a visually impaired girl who rediscovers her dreams in darkness. Built on gesture and voice input, the film incorporates four generative AI modalities—environment, sound, 3D models, and dialogue—enabling dynamic multimodal generation at key narrative points. Real-time feedback to user behavior and story progression supports a personalized narrative system, showcasing the creative and expressive potential of generative AI in VR storytelling.

CCS Concepts

• Human-centered computing → Virtual reality; • Applied computing → Arts and humanities.

Keywords

Cinematic Virtual Reality, Generative Artificial Intelligence, Interactive Storytelling, Immersive Experience

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1 INTRODUCTION

With the rapid advancement of virtual reality technology, cinematic storytelling is evolving from traditional linear structures toward spatial and interactive forms, transforming the audience’s role from passive viewers to active participants. Meanwhile, generative artificial intelligence, with its powerful capabilities across multiple modalities—including text, images, 3D models, and audio—is reshaping the digital content creation process [Banh and Strobel 2023], driving narrative expression toward greater immersion, dynamic responsiveness, and personalization [Hashim et al. 2023].

Currently, the integration of generative AI with VR films remains largely confined to the early stages of production, such as scriptwriting and asset generation. Only a few projects incorporate

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real-time generative AI into the audience interaction process [Hirzle et al. 2023]. TouchStar emerges within this context as a creative exploration, integrating multiple generative AI technologies. In the film, users' voice input and gesture interactions directly influence the audiovisual content, resulting in a dynamic and personalized narrative experience.

2 DESIGN: GENERATIVE AI INTEGRATION

In virtual reality films, the construction of an immersive narrative experience typically relies on the synergy of multiple elements, including the visual environment, auditory atmosphere, interactive objects, and narrative progression. To enhance this, the film integrates four generative AI modalities—environment textures, sound effects, 3D models, and voice dialogues—corresponding to scene construction, emotional tone, interaction response, and semantic feedback (see Figure1). These allow user input to dynamically influence scene content, enabling personalized interactive storytelling.

2.1 Environment Texture Generation

As the visual backdrop of a virtual space, the 360° environment texture significantly shapes the user's first impression of immersion [Teixeira et al. 2024]. In TouchStar, the system maps narrative and emotional states to control texture, color, and style, visualizing mood and enhancing the audience's perception of the character's state. For example, the accident scene features cold gray-blue tones and clouds. In contrast, the final scene shifts to warm orange hues and soft textures to convey hope.

2.2 Sound Effect Generation

Each scene is assigned emotional keywords to drive ambient sound generation in coordination with environment textures. Star-shaped particle effects serve as interactive cues; when triggered by hand gestures, the system generates audio feedback reflecting the protagonist's emotional state. For instance, slow piano melody in the first scene evokes oppression, while light violin tune in the fourth suggests hope. This interaction can be triggered repeatedly, enhancing emotional resonance and encouraging exploration.

2.3 3D Model Generation

To enhance creative participation and narrative connection, the film incorporates 3D model generation, turning personalized input into narrative props. In the second scene, viewers can customize a 3D celestial body by selecting parameters like terrain, atmosphere, and structure. The system generates the model in real time, which the audience then hands to the protagonist to advance the story as a narrative prop.

2.4 Voice Dialogue Generation

In the second scene, the film introduces a natural language-based voice dialogue module, allowing users to ask the agent questions. The system identifies intent through keyword and context analysis, generating real-time voice and text responses. Inquiries about recent astronomical events trigger narrative progression, while general science questions are answered via the AI's knowledge base, enhancing exploration and interactivity.

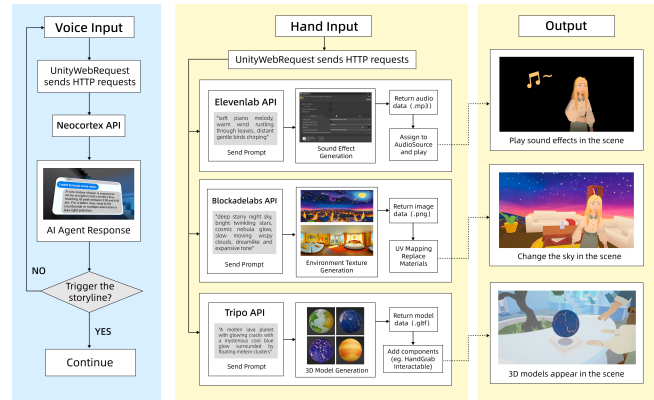


Figure 2: Technical Approach Implemented in the Film.

3 IMPLEMENTATION

The film is developed in Unity and deployed on Meta Quest 3. Generative AI modules are integrated via API, enabling the system to generate environment textures (Blockade Labs), sound effects (ElevenLabs), 3D models (Tripo), and voice dialogues (Neocortex) based on user voice and gesture input (see Figure2). Unity's event system parses interaction data and sends HTTP requests to external services. The returned multimodal results—audio, images, models, or text—are dynamically bound to scene objects at runtime, supporting real-time content rendering and narrative progression.

4 DISCUSSION AND FUTURE WORK

This project demonstrates the practical potential of generative AI in VR interactive storytelling. Preliminary user feedback suggests it enhances engagement and narrative immersion. In practice, certain generative processes exhibit response delays, which may impact immersion in high-frequency interaction scenarios. Future improvements could involve model optimization and local deployment. Our subsequent work will explore more efficient generation strategies and more natural interaction methods, aiming to expand the application of AI in nonlinear storytelling and personalized expression.

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